Youssef Kabbaj

MSc Applied Physics — machine learning for molecular and physical systems

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EPFL graduate: facilitated Swiss work authorisation
Available from April 2027

Profile

MSc student in Applied Physics at EPFL (graduating March 2027), working at the interface of **statistical physics** and **generative machine learning**. Currently developing **discrete diffusion models** for conditioned **protein sequence generation**; previously built transformer-parametrised **Potts models** (attention DCA) for protein families, and a deep-learning pipeline for ultrasound motion estimation and object localization in industry. Comfortable across the stack: stochastic processes and sampling theory, PyTorch training on GPU, and molecular simulation (GROMACS, CP2K, QM/MM).

Education

EPFL — Master of Science in Applied Physics 09/2024 – present
GPA 5.68 / 6 — expected March 2027

EPFL — Bachelor of Science in Physics 09/2021 – 07/2024
GPA 5.69 / 6

EPFL — Cours de Mathématiques Spéciales (CMS) 09/2020 – 07/2021
GPA 5.87 / 6

Groupe Scolaire Atlas, Morocco — Baccalauréat, Mathematical Sciences 07/2020
18.94 / 20

Technical skills

- **Statistical physics & stochastic modelling** — Langevin and Fokker–Planck equations, stochastic differential equations, Monte Carlo and MCMC sampling, maximum-entropy and Potts models, free-energy landscapes.
- **Physical simulation & numerical methods** — modelling physical problems from first principles and simulating them: Langevin and MD integrators, Monte Carlo, spectral and Fourier methods, ODE/PDE solvers, GPU simulation in PyTorch.
- **Generative modelling** — diffusion and score-based models, flow matching, autoregressive models, attention and transformers, energy-based models; training by maximum and pseudo-likelihood.
- **Machine learning** — supervised and reinforcement learning, synthetic data generation with ground truth, metric design for unlabelled data, model evaluation and failure analysis.
- **Molecular simulation & chemistry foundations** — set up and ran QM/MM and molecular-dynamics workflows in GROMACS, CP2K and MiMiC during a semester project; structure visualisation in PyMOL; bond types, intermolecular interactions, binding and affinity energies: enough to work alongside chemists and read structural data critically.
- **Programming** — Python (PyTorch, NumPy, scikit-learn), C++, MATLAB, Git; basic experience with Linux and running jobs on HPC clusters.

Research and industry experience

Laboratory of Statistical Biophysics, EPFL Ongoing, from 11/2026
Master thesis: Diffusion models for conditioned protein sequence generation and their dynamics

- Implement a discrete diffusion model for protein sequence generation.
- Study the dynamics of generation: speciation time and mode collapse.
- Condition generation on protein family and function.

Bracco Suisse S.A. 02/2026 – 08/2026
Industry internship: deep-learning pipeline for contrast-enhanced ultrasound image processing

- Built and trained two deep-learning models on ultrasound sequences: one for optical flow estimation, one for localisation.
- Wrote a dedicated synthetic data generator for each model, designed to reproduce the appearance and statistics of real acquisitions, so that both could be measured against known ground truth.

- On synthetic data, the optical-flow model reached a motion-correction error of roughly 5%, and the localisation model a precision and accuracy both above 95%.
- Designed reference-free metrics to quantify performance on real, unlabelled clinical data where no ground truth exists, and fed the observed failure modes back into the data generators.

Laboratory of Statistical Biophysics, EPFL 02/2025 – 07/2025
Summer internship and semester project: Biased generative protein models based on attention DCA

- Fitted a Potts model to a protein family, with the couplings parametrised by a factored self-attention layer and trained by reweighted pseudo-likelihood maximisation.
- Sampled new sequences with a Gibbs-like conditional sampler at controlled inverse temperature: they reproduce the natural one-site frequencies at $r = 0.99$ and the connected two-site correlations at $r = 0.87$ (train-versus-test ceiling 0.94), and are novel yet as diverse as natural sequences.
- Biased generation by conditioning on PCA coordinates, written exactly as a harmonic restraint on the Potts energy: sampling reaches any target sub-family, lifting an under-sampled one from 5% to 100% of chains while preserving the functional motif. Code: github.com/yk985/attention_dca_jdomains

Laboratory of Computational Chemistry and Biochemistry, EPFL 09/2025 – 11/2025
Semester project: QM/MM simulation of acetone — GROMACS, CP2K, MiMiC

- Ran QM/MM simulations of acetone in MiMiC and measured how different placements of the boundary between the quantum and classical regions change the computed energies and vibrational frequencies.

Swiss Plasma Center, EPFL 07/2024 – 01/2025
Semester project: Investigation of the Alfvén wave continuum in tokamaks*; preceded by a Summer-in-the-Lab research internship*

- Computed the Alfvén continuum spectrum of tokamak equilibria using Fourier analysis, and analysed the frequency gaps opened in it by the coupling between shear Alfvén and acoustic waves.

EPFL 09/2022 – present
Teaching assistant in analysis and electromagnetism (12 h/week)

Selected personal projects

Code and documentation: github.com/yk985

- **Generative models for protein sequences**
  Built a hierarchical Potts model as a synthetic ground truth with known statistics, then benchmarked autoregressive and diffusion generators against the exact target distribution.
  github.com/yk985/hierarchical-_Potts_models_learning
- **RNA reactivity prediction (Kaggle Ribonanza)**
  A transformer with 3.6 million parameters predicting per-nucleotide chemical reactivity from sequence, trained on about 750,000 noisy sequencing measurements. Re-using the low-quality data normally discarded as an auxiliary target, plus an auxiliary head predicting base-pairing probabilities, cut held-out error by 11% at equal compute.
- **Inference and control of stochastic dynamics**
  Two companion projects on noisy overdamped dynamics: recovering the drift field of a stochastic process from simulated trajectories, comparing five estimators from binned Kramers–Moyal to an amortised CNN that infers the whole field in one forward pass; and steering a microswimmer through a noisy low-Reynolds-number environment with reinforcement learning, benchmarked against the exact dynamic-programming optimum so that the policy gap is measured rather than assumed.
  github.com/yk985/drift_field_inference · github.com/yk985/microswimmer-rl
- **Fluctuation theorems on a pulled polymer**
  GPU Langevin simulation of a bead-spring chain dragged through a thermal bath by a moving trap (10^5 trajectories per protocol), testing the Jarzynski equality and the Crooks theorem against an exact free-energy reference. Mapped where the one-sided Jarzynski estimate breaks down (a $1.5\,k_BT$ bias for a 40-bead chain) and recovered the free-energy difference within its $\pm 0.04\,k_BT$ error bar with Bennett's acceptance ratio.
  github.com/yk985/Jarzynski_Crooks_Theorems

Languages

Arabic — mother tongue · French — C1 · English — C1
Willing to learn German for a role in German-speaking Switzerland.

Interests

Basketball and sport in general; contemplation and the larger questions about life. I will bring home-made tiramisu to the office.